Curriculum vitae

Last updated: 17th of August, 2026

Full name: Virinchi Venkata Rallabhandi

Website: https://virinchirallabhandi.github.io

Nationality: Australian

Qualifications:

- PhD in Mathematics from the University of Edinburgh, 2022 - 2026
  Thesis title: *A tale of spinors and positive energy theorems*
  Supervisor: James Lucietti
- Master of Advanced Study in Applied Mathematics from the University of Cambridge, i.e. Part III of the Mathematics Tripos, 2021 - 2022
- Master of Physics (theoretical physics stream) from the University of Western Australia (UWA), 2020 - 2021
- Bachelor of Science (Physics, Mathematics & statistics) from UWA, 2017 - 2019

Publications:

- V. Rallabhandi. Spinorial quasilocal mass for spacetimes with negative cosmological constant. *Advances in Theoretical and Mathematical Physics*, 30(6):1767-1819, 2026.
- V. Rallabhandi. On energy bounds in asymptotically locally AdS spacetimes. *Classical and Quantum Gravity*, 43(1):015019, 2026.
- B. Araneda, J. Lucietti and V. Rallabhandi. On toric self-dual Einstein gravitational instantons, 2026. Available at arXiv[math.DG/2606.21455].

Talks (past and upcoming):

- Universidad Autónoma de Madrid, "SIFTS" summer school - July 19, 2024
- Beijing Institute of Mathematical Sciences and Applications, general relativity seminar - October 17, 2025. Recording available at https://bimsa.net/bimsavideo.html?id=60887.mp4.
- University of Cambridge, mathematical physics seminar - November 4, 2025
- Australian Mathematical Society annual conference - December 9, 2025
- Edinburgh Mathematical Physics Group seminar - January 28, 2026
- Polish Academy of Sciences, Simons Programme on Twistor Theory and its Applications - September 8, 2026

Research experience:
My PhD studies focussed broadly on applications of differential geometry to problems in general relativity. The bulk of my work was concentrated on the theme of positive energy theorems and quasilocal mass. My main achievements were the proof of a positive energy theorem for asymptotically, locally AdS spacetimes with a certain class of boundary geometry, the first complete proofs of BPS inequalities in this setting and the development of a new definition of quasilocal mass for spacetimes with negative cosmological constant which satisfies a number of physically desirable properties. In the last year, I've tried to broaden my research to include purely differential geometry problems. In this direction, Bernardo Araneda, James Lucietti and I studied the classification problem for toric, self-dual, Einstein 4-manifolds, providing a classification for an infinite family of manifolds - including some familiar, textbook examples - characterised by possessing a conformally ALE extension. Other topics I studied during my PhD included some ongoing work exploring geometric flows of quasilocal masses and some work on static black hole uniqueness theorems.

Employment:

- During my PhD, I was employed by the University of Edinburgh as a teaching assistant for the courses, "*proofs and problem solving*," "*several variable calculus and differential equations*," "*fundamentals of pure mathematics*," "*honours analysis*," "*honours differential equations*," "*topics in mathematical physics A* [focussed on black holes]," "*introduction to mathematics at university*," "*classical field theory*" and "*gravity*" at various times. In a few cases, this also included preparing tutorial materials and occasionally lecturing.
- Between March 2020 and June 2021, I was employed by UWA to work as a tutor for the first year mathematics unit, "*mathematical theory and methods*."
- I was employed as a tutor for the 2021 Western Australian Mathematics Summer School (WAMSS). WAMSS is a week long residential school for promising final year secondary school students, aiming to introduce them to interesting areas of maths which they wouldn't otherwise be able to access at school.

Academic achievements during tertiary education:

- Equal 9th (3rd in Australia) in the pairs division (with Alexander Rohl) of the 2017 *Simon Marais Mathematics Competition* (an Asia-Pacific wide undergraduate maths competition modelled on the Putnam competition)
- Came 3rd, 1st and 3rd respectively in the 2017, 2018 and 2019 *Blakers Mathematics Competition* (a Western Australia wide undergraduate maths competition)
- Winner of the 2021 *Muriel and Colin Ramm Prize and Medal* (for the highest scoring student in the Master of Physics at UWA)
- Joint winner (with Jesse Schelfhout) of the 2019 *John de Laeter Medal* (for the highest scoring 3rd year undergraduate physics student at a Western Australian university) and winner of the 2019 *Lance Maschmedt Physics Prize* (for highest average score across the core 3rd year physics units at UWA)
- Winner of the 2019 *Ken Freeman Prize in Astrophysics and Space Science* (for highest score in the unit, PHYS3003, at UWA)
- Winner of the 2017 *Lady James Prize in Science* (for the highest score across the 1st year physics or chemistry units at UWA)

Programming languages:

- Proficient in *Java*
- Moderate knowledge of *Mathematica*
- Basic knowledge of *Python*

Funding:

- School of Mathematics Studentship - for my PhD at Edinburgh
- Part III International Scholarship - for my master's at Cambridge
- Muriel and Colin Ramm Postgraduate Scholarship in Physics - for my master's at UWA

Referees:

- Professor James Lucietti (j.lucietti@ed.ac.uk)
- Dr Jan Sbierski (jan.sbierski@ed.ac.uk)
- Dr Tim Adamo (t.adamo@ed.ac.uk)

Miscellaneous:

- Member of the Australian Labor Party
- Member of the Australian Republic Movement
- Life member of the Cambridge Union Society
- Long time tuba player